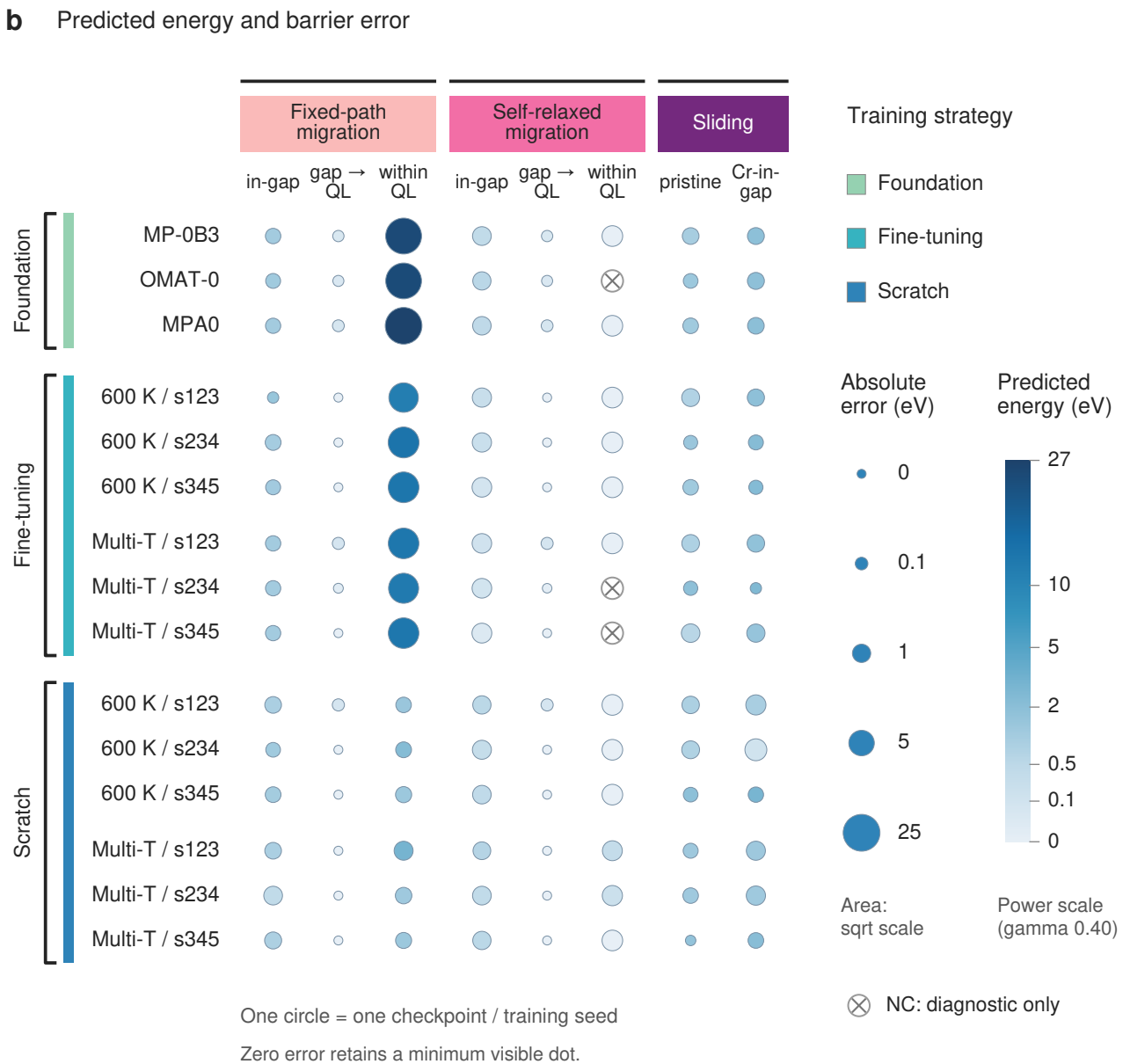
b
Predicted energy and barrier error
Fixed-path migration
Self-relaxed migration
Sliding
in-gap
gap → QL
within QL
in-gap
gap → QL
within QL
pristine
Cr-in-gap
Foundation
MP-0B3
OMAT-0
MPA0
Fine-tuning
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Scratch
600 K / s123
600 K / s234
600 K / s345
Multi-T / s123
Multi-T / s234
Multi-T / s345
Training strategy
Foundation
Fine-tuning
Scratch
Absolute error (eV)
0
0.1
1
5
25
Predicted energy (eV)
27
10
5
2
0.5
0.1
0
Area: sqrt scale
Power scale (gamma 0.40)
NC: diagnostic only
One circle = one checkpoint / training seed
Zero error retains a minimum visible dot.